UNIT 1 · KINEMATICS · WEEK 3 · LAB DAY 2 OF 2

Cart Lab — Day 2: Analyze & Draft

Physics 2026–27 · Tuesday, October 13 · Lab Report I due Thursday, October 15

NAME _____ GROUP _____

Today's job is the meaning. If acceleration is constant, $x = \frac{1}{2}at^2$, so x plotted against t^2 should be a straight line with slope $\frac{1}{2}a$. Test whether that model fits your data.

1 · Finish the table

FROM YOUR DAY 1 SHEET · 10 MIN

Copy your positions and average times. The ± column is half the range of the three trials: `(longest − shortest) / 2`. Then square each average time, keeping three significant figures at most.

POSITION X (M)	AVERAGE T (S)	± T (S)	T^2 (S^2)
0	0	—	0

2 · Straighten the curve

X VS T² · 20 MIN

Plot t^2 **across** and **position up**; label both axes with units and include `(0, 0)`. Then rule **one best-fit straight line** that passes *near* the points. Do not connect the dots.



3 · Slope → acceleration

UNITS THE WHOLE WAY · 10 MIN

a. Pick **two points on your line**, far apart — on the line, not data points. Write both with units.

b. Compute the slope, `rise / run`, with its unit.

c. Compare $x = \frac{1}{2}at^2$ with $y = mx$: the slope is $\frac{1}{2}a$, so $a = 2 \times \text{slope}$. Find a.

d. Rule the **steepest** and **shallowest** lines that still fit. Find a for each; ± is half the difference.

WORK

Result: $a =$ _____ ± _____ m/s^2

4 · What the data says

CONCLUSION DRAFT · 15 MIN

a. Is your x–t^2 graph straight? Does the line pass through the origin? If it misses, why might it?

b. Name **two** sources of uncertainty, and say which way each pushes your a — too high or too low.

c. On your Day 1 graph, draw a **smooth curve** through the x–t points. The report needs both graphs.

ANSWERS

Conclusion starter: The x–t^2 graph was _____, which _____ the hypothesis that the acceleration was constant. From the slope, $a =$ _____ ± _____ m/s^2.

5 · Peer review

SWAP BOTH SHEETS WITH ANOTHER GROUP · CHECK ONLY WHAT IS TRUE · 10 MIN

- ☐ Headers: variable and unit
- ☐ Cells: one number, no units
- ☐ Every time to 0.01 s
- ☐ Both graphs: axes labeled with units
- ☐ Data points are visible dots
- ☐ x–t smooth curve; x–t^2 one ruled line
- ☐ Slope has a unit, from points on the line
- ☐ a is stated with a ±

Reviewed by _____ One thing to fix: _____

Lab Report I, due Thu 10/15: *Question, Hypothesis, Materials, Procedure, Data, Analysis, Conclusion.*